



Technical Service Bulletin

Technical Service Bulletin: TSB120285	Released Date: 07-Nov-2018
Belt Shredding on QSX15 CM570 With Fan Centers 18" and Above	

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Core Issue

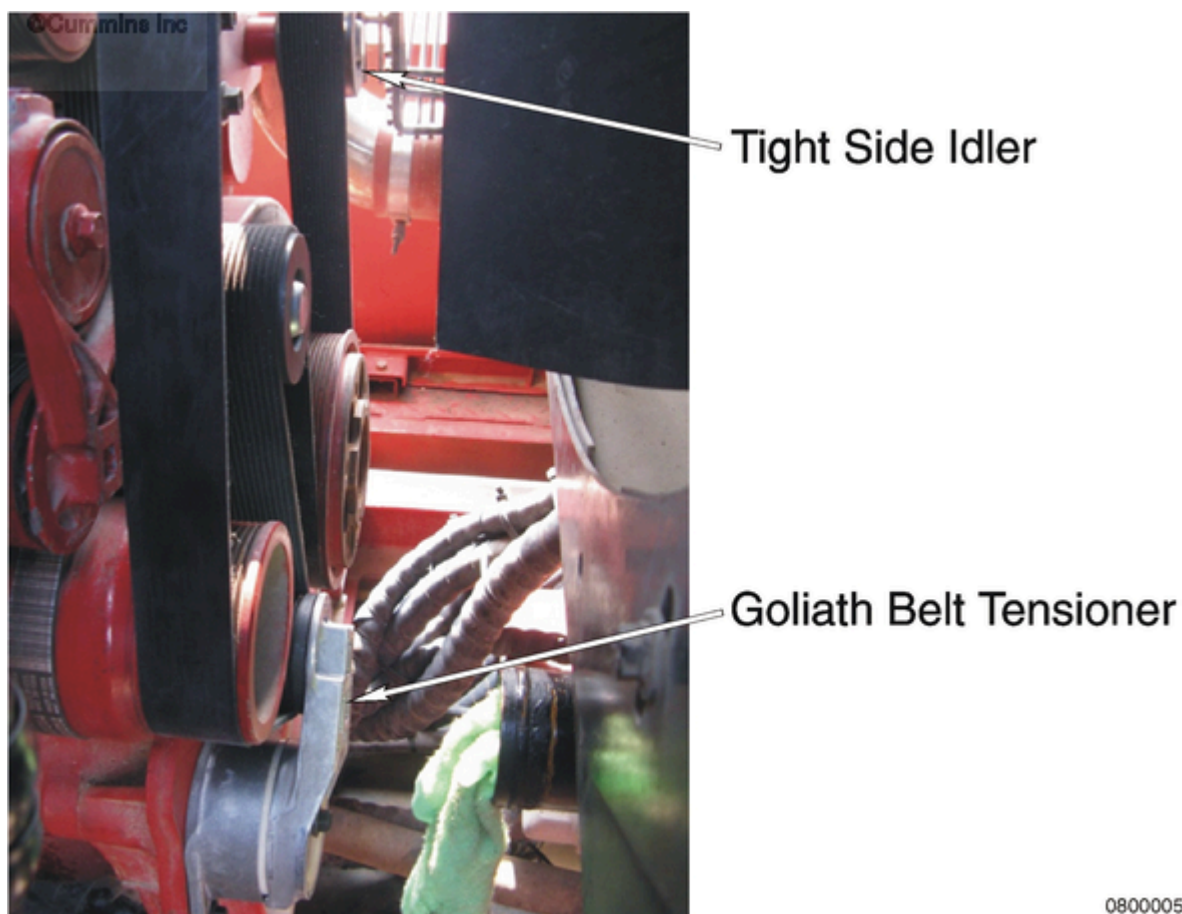
There are QSX engines with fan centers 18" and above experiencing belt shredding due to long belt spans between the fan hub and the crankshaft, as well as the fan hub and the water pump. The span lengths are susceptible to vibration and quick deceleration events which cause the belt to jump grooves and start shredding on the edges of the pulleys. The customer will notice the coolant temperature rising, and find that the belt is missing ribs or is completely gone.

Confirmation

To determine if the belt shredding on the unit is this issue, measure the fan center between the crankshaft and the fan hub. Compare the shredded belt to photographs 3, 4, and 5 located at the end of this document to determine if it is a match.

Resolution

If the belt shredding and the fan center match the verification, a tight side idler needs to be installed on the unit to reduce the amount of vibration seen during deceleration events. Addition of a Goliath tensioner, on some FA options, increases the tension in the system, reducing belt vibration.



Photograph 1: Side View of Goliath Tensioner Configuration and Tight Side Idler

The following parts need to be ordered to install the Goliath belt tensioner configuration and the tight side idler:

Table 1: Goliath Belt Tensioner Configuration Components		
Part Number	Part Description	Quantity Required
See Table 2 for belt part number	Belt	1
3104029	Belt Tensioner	1
3093936	Capscrew, M10 x 90 mm	3
3902460	Capscrew, M10 x 25 mm	1
3681715	Pilot Adapter	1
70760	Dowel Pin	1
3026269	Washer	1
3681587	Idler Pulley	1
3914407	Capscrew, M10 x 100 mm	1
3935013	Dust Shield	1
3681191	Belt Tensioner Bracket	1
3681390	Capscrew, M10 x 150 mm	1
3009330	Washer	1
Tight Side Idler Components		

Table 1: Goliath Belt Tensioner Configuration Components		
3681587	Idler Pulley	1
3035803	Hex Nut	2
3688979	Idler Pulley Bracket	1
3093940	Capscrew, M12 x 45 mm	2
3935013	Dust Shield	1
3903112	Capscrew, M10 x 30 mm	1
3026269	Washer	1

Table 2: Belt Selection for FA options over 18 inches		
Factory Installed FA Option	Factory Installed Fan Center Height	Goliath Configuration Belt Part Number
FA1545	19 inches	3103838
FA1533	19 inches	3103835
FA1539	20 inches	3103837
FA1541	20 inches	3103838

Table 3: Belt Selection with Tight Side Idler Only		
Factory Installed FA Option	Factory Installed Fan Center Height	Orion Belt Part Number
FA1541	20 inches	3100915
FA1677	20 inches	3100915
FA1678	25 inches	3100269

All service procedures referenced below are in Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239.

Removal of fan drive components

- Remove the fan drive belt, belt tensioner, and belt tensioner support from the peg board.
 - Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-002-tr.html>)
 - Refer to Procedure 008-080 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>)

Installation of the tight side idler:

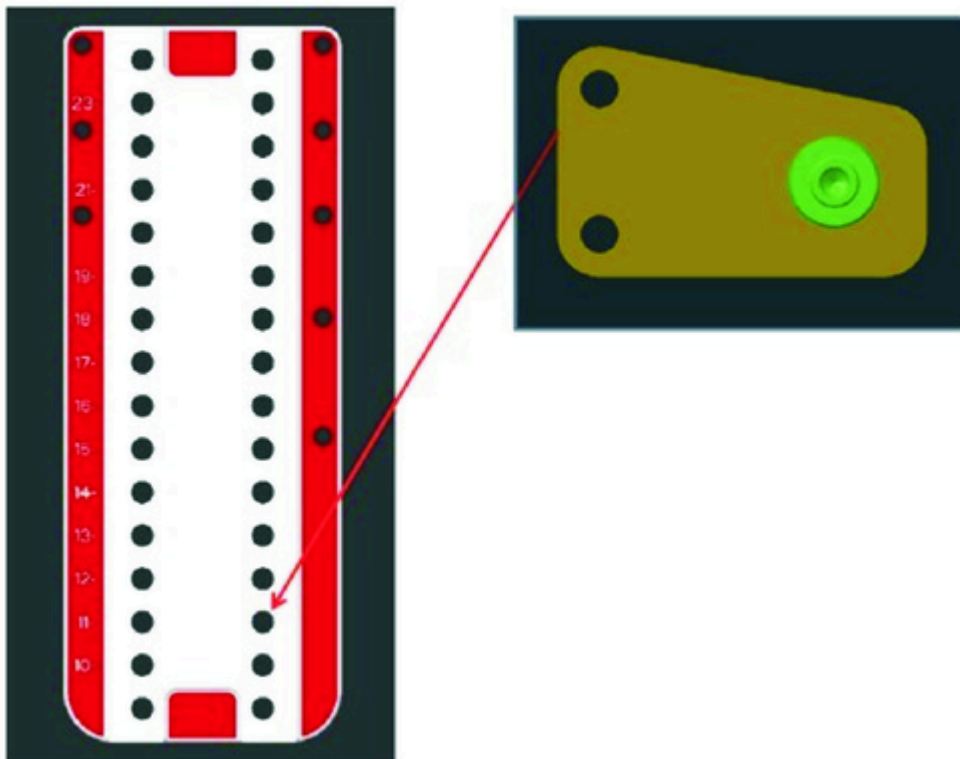
- Remove the fan hub support.
 - Refer to Procedure 008-088 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-088-tr.html>)
- Install the idler pulley to the idler pulley bracket. Use capscrew, Part Number 3903112, and washer, Part Number 3026269.
 - Tighten the capscrew to 45 N•m [33 ft-lb].
- Install the dust shield. Refer to Procedure 008-030 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-030-tr.html>)

- Install the idler bracket to the fan hub support at the 2nd and 4th mounting holes on the right side of the board. Use hex nuts, Part Number 3035803, and capscrews, Part Number 3093940.
 - Tighten the capscrews to 75 N•m [55 ft-lb].
- Install the fan hub support.
 - Refer to Procedure 008-088 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-088-tr.html>)

Location of tight side idler bracket (no Goliath tensioner installed):

- Top hole of the tight side idler bracket, per Illustration 1, goes into the right side number eleven hole of the peg board for both options.

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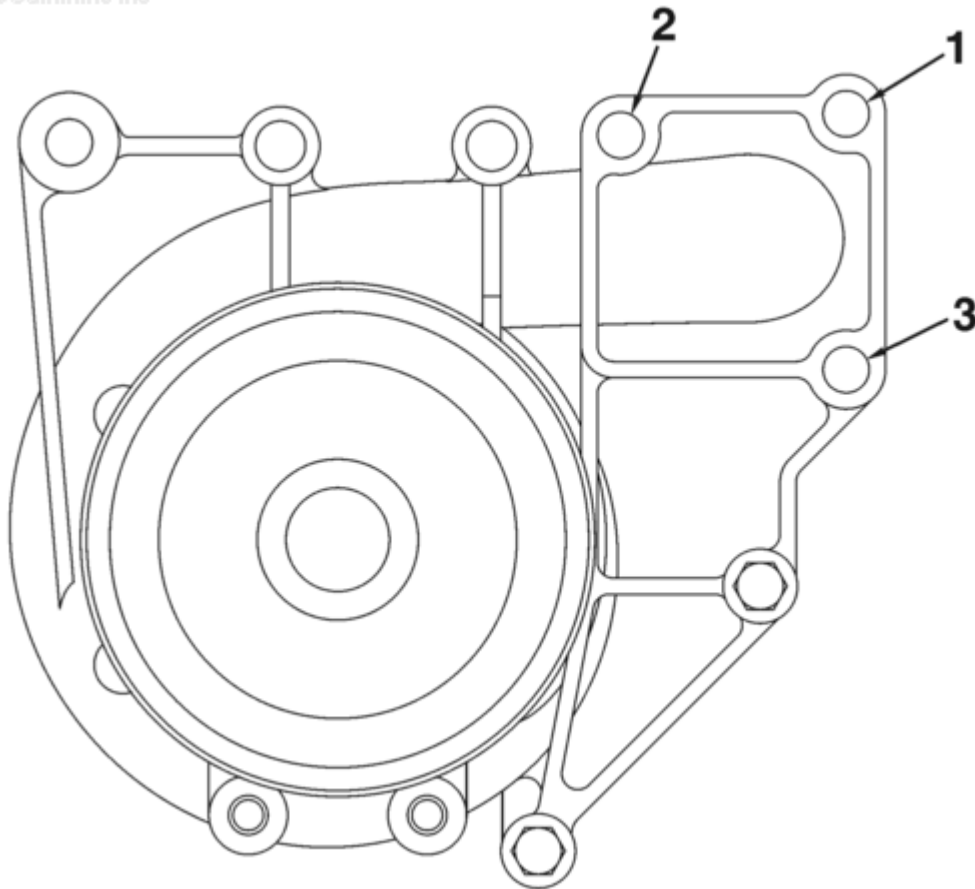


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Illustration 1 - Tight Side Idler Bracket

Installation of the Goliath belt tensioner configuration:

- Remove the 3 capscrews on the right side of the water pump, as identified in Illustration 2.



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Illustration 2 - Capscrews to be Removed

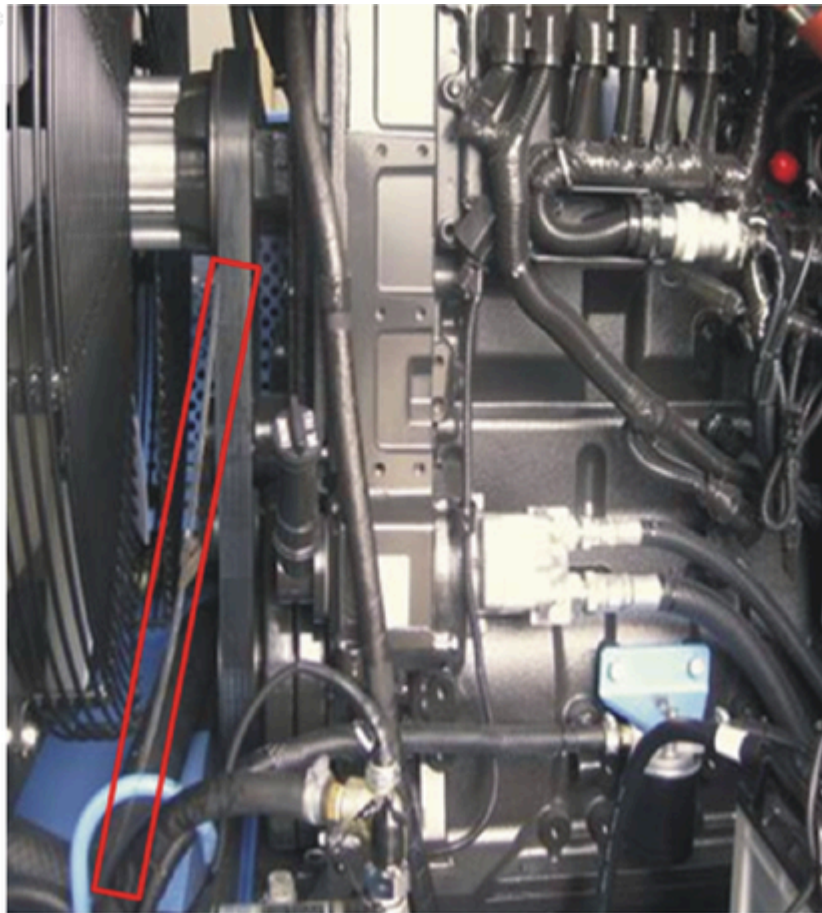
- Install the belt tensioner bracket, Part Number 3681191, where the three capscrews were removed from the water pump. Per Illustration 2, capscrew 1 is Part Number 3681390, capscrews 2 and 3 are Part Number 3093936. Capscrew 3 requires washer, Part Number 3009330.
 - All capscrews are to be tightened to 58 N•m [43 ft-lb].
- Install the pilot adapter, Part Number 3681715, and the idler pulley, Part Number 3681587.
 - Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-062-tr.html>)
- Install the Goliath belt tensioner (low mount-large tensioner).
 - Refer to Procedure 008-080 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>)
- Install the fan drive belt.
 - Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-002-tr.html>)



Photograph 3 - Rib missing while belt is installed on the fan hub pulley.



Photograph 4 - The eleventh and twelfth ribs separating from the belt.



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Photograph 5 - The ribs separating from the belt shown in the red box.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2012-11-6	Module Created
2014-9-8	none
2018-10-19	Removed FA1542 references.

Last Modified: 07-Nov-2018
